IIOT MIDTERM PROJECT REPORT

Smart Surveillance & Intelligent Street Lighting IoT System Report

1. Aim

The objective of this project is to design, model, and simulate an automated, intelligent urban ecosystem within Cisco Packet Tracer. The system integrates advanced IoT-enabled infrastructure components, specifically smart surveillance networks, environmental hazard safeguards, adaptive resource management systems, and smart responsive lighting solutions. By applying real-time sensor-triggered logic, the system seeks to maximize municipal energy efficiency, enhance community safety during nighttime hours, provide robust automated emergency responses, and preserve urban public resources without requiring manual administrative intervention.

2. Problem Statement

Modern municipal areas encounter significant resource mismanagement and delayed hazard responses due to traditional, unlinked infrastructure. Streetlights typically operate on fixed timers, wasting excessive amounts of electricity during periods of zero pedestrian or vehicular activity. Security networks operate in passive formats, failing to actively signal surrounding safety components when motion or unauthorized entries occur. Furthermore, localized safety hazards like urban fires or water tank overflows often require manual, delayed operator mitigation.

To solve these critical urban operational challenges, an automated infrastructure framework must be built to instantly process sensor readings and coordinate multi-component responses.

Figure 2.1: Inefficient street lighting power grid energy bleed.

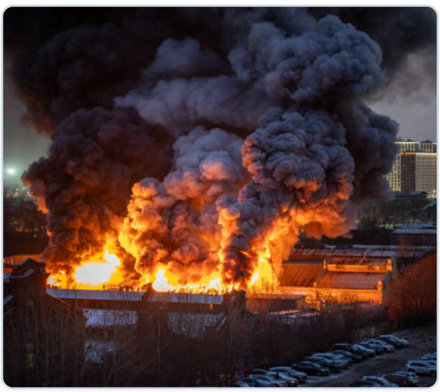


Figure 2.2: Infrastructure response delays during fire safety risks.

Figure 2.3: Utility water asset overflow from lack of automated control.

3. Scope of the Solution

The solution addresses four core areas of urban automation:

- **Dynamic Security and Surveillance Networking:** Implements synchronized motion tracking that instantly activates webcams and secondary high-intensity illumination lamps whenever unexpected physical activity occurs in designated zones.
- **Automated Environmental & Fire Protection:** Deploys specialized smoke detection arrays that automatically handle containment via variable-speed ventilation blowers, trigger audible municipal sirens, and engage localized fire suppression sprinklers.
- **Resource-Efficient Utility Controls:** Features smart fluid monitoring systems to regulate water storage levels and eliminate pump dry-running or waste overflows through real-time telemetry.
- **Renewable Power Integration:** Incorporates solar generation and centralized battery storage banks to maintain sustainable power loops for all embedded town sensors.

4. Overview/Architecture of the Solution

The system relies on a central intelligent IoT Home/Smart Gateway that coordinates all communications using an instantaneous conditional rule engine. Edge network devices constantly post operational telemetry to the central gateway, which checks conditions against a pre-programmed truth table to execute corresponding action matrices.

Key Inter-Device Logical Architecture:

- **Security Loop 1 & 2:** Motion Detectors route logic to activate specific target cameras (Cam1 / Cam2) and ambient streetlights (LAMP1 / LAMP2) on-demand, turning off when motion ceases to prevent energy bleed.
- **Environmental Containment Matrix:** The Smoke Detector sets linear tiers for air purification. Low smoke triggers low-volume extraction (Blower = Low), whereas severe levels spark full ventilation (Blower = High), audible warnings, and water suppression systems.

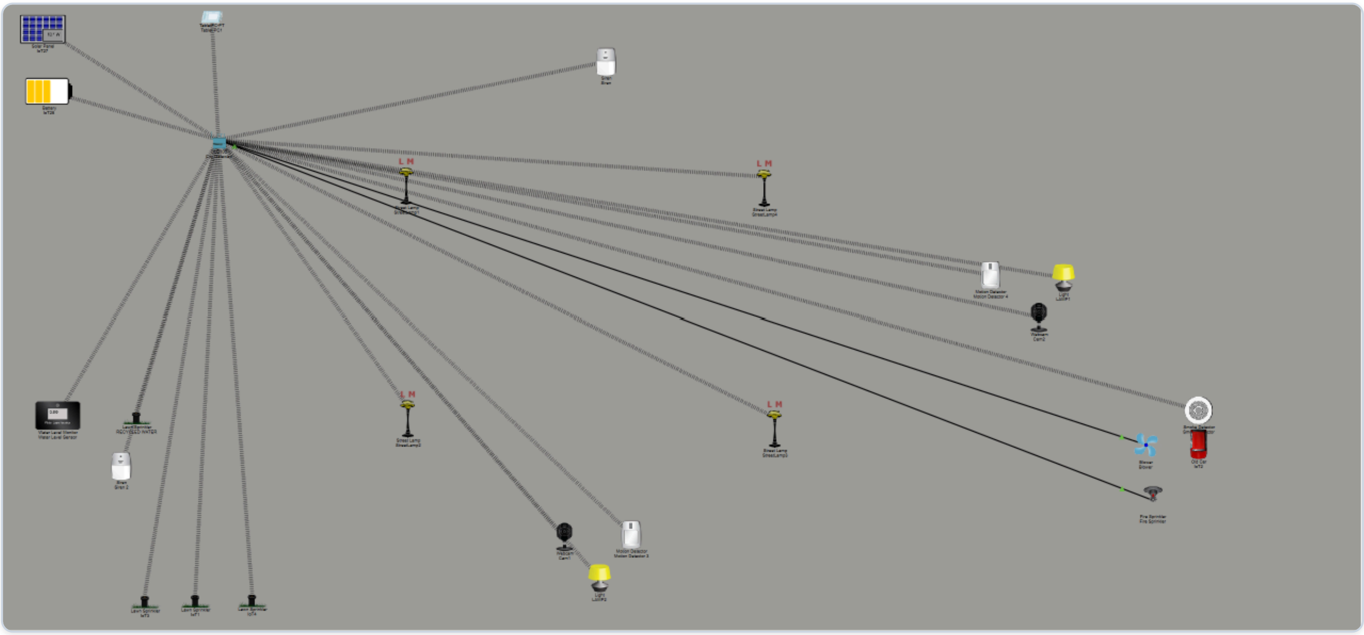
5. Required Components to Develop Solutions

Component Category	Device Name (ID)	Primary Operational Functionality
Central Core	Home Gateway / IoT Server	Establishes local wireless/wired networking, registers nodes, and runs the rule engine.
Lighting Nodes	Street Lamps (1 to 4), LAMP1, LAMP2	Provides localized adaptive illumination based on localized sensing loops.
Surveillance	Motion Detectors (3 & 4), Webcams (1 & 2)	Tracks motion events and captures live visual verification feeds.
Hazards & Safety	Smoke Detector, Blower, Fire Sprinkler, Siren	Identifies toxic fumes, runs variable extraction, and activates physical fire suppression.

Component Category	Device Name (ID)	Primary Operational Functionality
Resource Monitoring	Water Level Sensor, Lawn Sprinklers, Siren 2	Maintains automated liquid asset delivery and overflow alerts.
Power Grid	Solar Panel, Battery Bank	Provides off-grid renewable power generation and storage.

6. Simulated Circuit (Screenshot of the Logical System View)

The topological architecture maps out localized smart infrastructure grids that are linked to the primary distribution gateway. The layout ensures reliable connectivity across municipal zones.



Automated Conditional Execution Rules

The IoT system operates precisely under the following conditional truth metrics configured within the central server instance:

Rule Name	Triggering Condition	Automated System Responses
Night Security 1_ON	Motion Detector 4 On is true	Set LAMP1 Status to On; Set Cam2 On to true.
Night Security 2_ON	Motion Detector 3 On is true	Set Cam1 On to true; Set LAMP2 Status to On.
Recycled Water System_Pump_ON	Water Level Sensor Water Level ≥ 10.0 cm	Set Siren 2 On to true.

Rule Name	Triggering Condition	Automated System Responses
Smoke_Alarm	Smoke Detector Level ≥ 0.168804	Set Siren On to true; Set Fire Sprinkler Status to true; Set Blower Status to High.
Recycled Water System_Pump_Off	Water Level Sensor Water Level < 10.0 cm	Set Siren 2 On to false.
Night Security 2_OFF	Motion Detector 4 On is false	Set Cam1 On to false; Set LAMP2 Status to Off.
Night Security 1_OFF	Motion Detector 3 On is false	Set Cam2 On to false; Set LAMP1 Status to Off.
Smoke_Alarm_OFF	Smoke Detector Level < 0.168804	Set Siren On to false; Set Fire Sprinkler Status to false.
Small_Smoke	Smoke Detector Level > 0.1	Set Blower Status to Low.
Small_Smoke_OFF	Smoke Detector Level < 0.1	Set Blower Status to Off.

IoT Monitor

IoT Server - Device Conditions

Home | Conditions | Editor | Log Ou

Actions	Enabled	Name	Condition	Actions
Edit Remove	Yes	Night Security 1_ON	Motion Detector 4 On is true	Set LAMP1 Status to On Set Cam2 On to true
Edit Remove	Yes	Night Security 2_ON	Motion Detector 3 On is true	Set Cam1 On to true Set LAMP2 Status to On
Edit Remove	Yes	Recycled Water System_Pump_ON	Water Level Sensor Water Level >= 10.0 cm	Set Siren 2 On to true
Edit Remove	Yes	Smoke_Alarm	Smoke Detector Level >= 0.168804	Set Siren On to true Set Fire Sprinkler Status to true Set Blower Status to High
Edit Remove	Yes	Recycled Water System_Pump_Off	Water Level Sensor Water Level < 10.0 cm	Set Siren 2 On to false
Edit Remove	Yes	Night Security 2_OFF	Motion Detector 4 On is false	Set Cam1 On to false Set LAMP2 Status to Off
Edit Remove	Yes	Night Security 1_OFF	Motion Detector 3 On is false	Set Cam2 On to false Set LAMP1 Status to Off
Edit Remove	Yes	Smoke_Alarm_OFF	Smoke Detector Level < 0.168804	Set Siren On to false Set Fire Sprinkler Status to false
Edit Remove	Yes	Small_Smoke	Smoke Detector Level > 0.1	Set Blower Status to Low
Edit Remove	Yes	Small_Smoke_OFF	Smoke Detector Level < 0.1	Set Blower Status to Off

7. Execution Video of the Demo

The interactive operational dynamics, sensor triggers, and immediate response loops of this intelligent urban monitoring system have been captured in the live demonstration placeholder link below.

Video Demo Link:
https://drive.google.com/file/d/1GpWMRuJEjEWKxouVnFhFO5psrcR-vGEQ/view?usp=drive_link
Github repository link:

https://github.com/vinayaksanthosh2024-debug/VIT-AIIOT_PROJECT-24BEC0190_Vinayak/blob/main/IIOT%20PROJECT%20REPORT.pdf